MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

ASSIGNMENT - 1

Computational Methods and Stochastic Processes / MAT 5128

Due date: September 11, 2025 Maximum Marks: 6

Question 1 (3 marks)

Consider a graph G with vertex set

$$V(G) = \{s, t, x, y, z\}.$$

Let $W(u, v)$ denote the weight on the directed edge from the vertex u to vertex v. Suppose G is such that

$$\begin{array}{lll} W(t,y) = 1, & W(y,t) = 2, & W(t,x) = 1, \\ W(s,t) = 9, & W(s,y) = 4, & W(x,z) = 3, \\ W(z,x) = b+2, & W(y,z) = a+3, & W(z,s) = 6, \\ W(y,x) = 8. & & \end{array}$$

Here a is the last digit of your registration number and b is the last but one digit of your registration number. If both are zero take $a = 9$ and $b = 4$.

1. Draw the directed graph G.
2. Apply Dijkstra's algorithm to find the shortest paths from s to all the other vertices of G.

Question 2 (3 Marks)

Find a singular value decomposition of the matrix

$$A = \begin{bmatrix} a & 0 & a \\ 0 & 1 & 0 \\ b & 0 & b \end{bmatrix},$$

where a is the last digit of your registration number and b is the last but one digit of your registration number. If both are zero take $a = 9$ and $b = 4$.